S1. Composed MLV Machine Skeleton

The composed MLV machine is obtained by including the three verified component machines and synchronising their shared events under the interleaving-with-synchronization operator |||. The **includes** keyword declares the component machines and the **synchronises** keyword binds their shared events into system-level events. A synchronised event executes only when every participating component event is enabled; its guard is the conjunction of the participating guards and its action updates all participating machines jointly. Events local to a single component are interleaved without synchronisation.

```
machine Composed_MLV

includes
  GUI_Machine        as gui
  Alarms_Machine     as alrm
  Controller_Machine as ctrl

sees
  GUI_Context
  Alarms_Context
  Controller_Context

events
  Event INITIALISATION
    synchronises
      gui.INITIALISATION
      alrm.INITIALISATION
      ctrl.INITIALISATION
  end

  Event PowerOn
    synchronises
      gui.PowerOn
      ctrl.PowerOn
  end

  Event AlarmActivate
    synchronises
      alrm.ActivateBatteryAlarm
      gui.AlarmActivate
      ctrl.AlarmActivate
  end

  ...
end
```

Listing 1. Skeleton of the composed MLV machine.

S2. Results for the 5-5-5 and 30-30-30 Initial Sets

The main paper reports results for the 10-10-10 initial set (10 single-threat, 10 double-threat, and 10 triple-threat faulty composed systems, yielding 900 initial traces). This section provides the complete numerical results and fitted curve parameters for the 5-5-5 (450 initial traces) and 30-30-30 (2 700 initial traces) initial sets. The acquisition checkpoints, hidden evaluation set, classifier settings, and query strategies are identical to those described in the main paper.

2.1. Classifier Comparison (Entropy Sampling)

Table 1 reports grouped internal-validation F1-Macro under Entropy sampling for both the 5-5-5 and 30-30-30 initial sets across all three tasks and all six checkpoints. The classifier ordering matches the 10-10-10 results in the main paper at every row: the three boosting models lead, Random Forest is a distant fourth, and Logistic Regression is last.

Table 1. Classifier comparison under Entropy sampling, 5-5-5 and 30-30-30 initial sets: grouped internal-validation F1-Macro (%) at each checkpoint, all tasks.

Init. Set	Task	Classifier	Init	1%	2%	4%	8%	16%
5-5-5	A	Gradient Boosting	79.25	83.87	87.26	89.83	91.11	91.31
		LightGBM	80.16	84.36	87.47	89.93	91.10	91.28
		XGBoost	79.24	83.05	85.90	88.15	89.23	89.39
		Random Forest	47.86	51.84	54.79	57.11	58.23	58.29
		Logistic Regr.	22.17	25.33	27.70	29.56	30.45	30.60
	B	LightGBM	80.02	83.38	85.89	87.86	88.81	88.95
		XGBoost	77.93	82.14	85.29	87.76	88.95	89.14
		Gradient Boosting	77.12	81.14	84.09	86.31	87.42	87.59
		Random Forest	50.30	53.02	55.05	56.64	57.41	57.42
		Logistic Regr.	16.47	19.41	21.61	23.35	24.18	24.32
	C	LightGBM	76.35	80.80	84.10	86.70	87.94	88.14
		Gradient Boosting	76.36	80.07	82.79	84.83	85.86	86.03
		XGBoost	74.79	78.63	81.51	83.78	84.87	85.03
		Random Forest	41.20	44.93	47.69	49.86	50.91	50.96
		Logistic Regr.	8.36	11.57	13.97	15.86	16.77	16.92
30-30-30	A	Gradient Boosting	87.25	90.07	91.32	91.94	92.02	92.04
		LightGBM	88.16	90.36	91.44	91.93	92.00	91.91
		XGBoost	87.24	88.87	89.68	90.06	90.11	90.13
		Random Forest	55.86	57.73	58.64	59.06	59.01	59.03
		Logistic Regr.	30.17	30.80	31.13	31.28	31.31	31.33
	B	XGBoost	85.43	87.96	89.21	89.77	89.85	89.87
		LightGBM	87.52	88.74	89.35	89.63	89.67	89.58
		Gradient Boosting	84.62	86.82	87.77	88.24	88.31	88.33
		Random Forest	57.80	58.04	58.17	58.23	58.14	58.16

Continued on next page

Table 1 (continued)

Init. Set	Task	Classifier	Init	1%	2%	4%	8%	16%
		Logistic Regr.	23.97	24.56	24.86	25.00	25.03	25.05
	C	LightGBM	84.85	87.15	88.27	88.78	88.85	88.77
		Gradient Boosting	84.86	86.01	86.45	86.70	86.74	86.76
		XGBoost	83.29	84.68	85.38	85.70	85.75	85.77
		Random Forest	49.70	50.89	51.48	51.74	51.68	51.70
		Logistic Regr.	16.86	17.28	17.50	17.61	17.63	17.65

2.2. Query-Strategy Comparison (LightGBM Fixed)

Table 2 reports hidden-test F1-Macro with LightGBM fixed, for both alternative initial sets across all three tasks. Entropy and Margin lead Random and Jaccard Diversity in every row, consistent with the 10-10-10 results reported in the main paper.

Table 2. Query-strategy comparison using LightGBM, 5-5-5 and 30-30-30 initial sets: hidden-test F1-Macro (%) at each checkpoint, all tasks.

Init. Set	Task	Strategy	1%	2%	4%	8%	16%
5-5-5	A	Margin	83.47	86.66	89.09	90.35	90.58
		Entropy	83.46	86.64	89.18	90.43	90.65
		Random	82.11	84.24	86.03	86.93	87.10
		Jaccard Diversity	81.99	84.18	85.93	86.81	86.98
	B	Entropy	82.40	84.98	87.03	88.06	88.24
		Margin	81.81	83.94	85.54	86.40	86.56
		Random	81.74	83.76	85.46	86.33	86.49
		Jaccard Diversity	81.65	83.75	85.43	86.29	86.45
	C	Entropy	79.75	83.12	85.80	87.12	87.36
		Margin	78.74	81.37	83.36	84.40	84.60
		Jaccard Diversity	78.53	81.08	83.12	84.14	84.33
		Random	78.19	80.36	82.17	83.09	83.26
30-30-30	A	Margin	89.88	90.93	91.50	91.65	91.68
		Entropy	89.86	91.01	91.58	91.70	91.61
		Random	87.73	87.93	88.08	88.18	88.23
		Jaccard Diversity	87.68	87.84	87.96	88.06	88.12
	B	Entropy	88.16	88.84	89.20	89.32	89.27
		Margin	87.25	87.38	87.54	87.64	87.70
		Random	87.08	87.31	87.47	87.57	87.63
		Jaccard Diversity	87.07	87.28	87.43	87.53	87.59
	C	Entropy	86.50	87.69	88.28	88.43	88.39
		Margin	84.96	85.30	85.56	85.67	85.73

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Table 2 (continued)

Init. Set	Task	Strategy	1%	2%	4%	8%	16%
		Jaccard Diversity	84.69	85.07	85.30	85.41	85.47
		Random	84.05	84.15	84.24	84.34	84.39

2.3. Fitted Saturating-Exponential Curve Parameters

Tables 3 and 4 report the parameters of the saturating-exponential curves

$$y(x) = y_0 + \Delta_{\text{fit}}\left(1 - e^{-kx}\right),$$

fitted to the six measured learning-curve points (initial point and the 1%, 2%, 4%, 8%, 16% checkpoints). The saturation point at 95% of the fitted improvement is $x_{95} = \ln(20)/k$. Fitted plateaus are within 1–2 points of the 10-10-10 values reported in the main paper. The fitted rate k is smallest at 5-5-5 and largest at 30-30-30, so x_{95} decreases from roughly 5.8–5.9% at 5-5-5 to 3.2–3.6% at 30-30-30 for the boosting models.

Table 3. Fitted curve parameters, classifier comparison (Entropy sampling), 5-5-5 and 30-30-30 initial sets, all tasks. Metric: grouped internal-validation F1-Macro (%).

Init. Set	Task	Classifier	Plateau (%)	k	x_{95} (%)	R^2
5-5-5	A	Gradient Boosting	91.33	0.518	5.78	0.9990
		LightGBM	91.32	0.510	5.87	0.9989
		XGBoost	89.43	0.507	5.90	0.9989
		Random Forest	58.38	0.516	5.81	0.9988
		Logistic Regr.	30.62	0.507	5.91	0.9988
	B	XGBoost	89.18	0.508	5.90	0.9989
		LightGBM	88.98	0.509	5.89	0.9989
		Gradient Boosting	87.61	0.520	5.76	0.9989
		Random Forest	57.50	0.517	5.79	0.9986
		Logistic Regr.	24.34	0.506	5.92	0.9989
	C	LightGBM	88.18	0.510	5.87	0.9989
		Gradient Boosting	86.04	0.519	5.77	0.9989
		XGBoost	85.07	0.507	5.91	0.9988
		Random Forest	51.04	0.516	5.80	0.9988
		Logistic Regr.	16.95	0.507	5.91	0.9989
30-30-30	A	Gradient Boosting	92.05	0.909	3.30	0.9997
		LightGBM	91.99	0.903	3.32	0.9982
		XGBoost	90.14	0.867	3.45	0.9990
		Random Forest	59.06	0.933	3.21	0.9975
		Logistic Regr.	31.32	0.830	3.61	0.9988
	B	XGBoost	89.88	0.881	3.40	0.9990

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Table 3 (continued)

Init. Set	Task	Classifier	Plateau (‰)	k	x_{95} (‰)	R^2
		LightGBM	89.65	0.906	3.31	0.9968
		Gradient Boosting	88.33	0.917	3.27	0.9998
		Random Forest	58.18	1.171	2.56	0.9495
		Logistic Regr.	25.04	0.833	3.60	0.9990
	C	LightGBM	88.84	0.904	3.31	0.9984
		Gradient Boosting	86.75	0.932	3.21	0.9999
		XGBoost	85.77	0.862	3.48	0.9990
		Random Forest	51.73	0.954	3.14	0.9962
		Logistic Regr.	17.64	0.804	3.73	0.9989

Note on Random Forest at 30-30-30, Task B. The poor fit ($R^2 = 0.9495$) for Random Forest on Task B at the 30-30-30 initial set is driven by a very small fitted gain (0.38 points): the large initial set already places Random Forest close to its low plateau before any candidate is added, leaving little room for the exponential curve to distinguish the growth phase.

Table 4. Fitted curve parameters, query-strategy comparison (LightGBM fixed), 5-5-5 and 30-30-30 initial sets, all tasks. Metric: hidden-test F1-Macro (%).

Init. Set	Task	Strategy	Plateau (‰)	k	x_{95} (‰)	R^2
5-5-5	A	Entropy	90.66	0.510	5.87	0.9992
		Margin	90.57	0.516	5.80	0.9992
		Random	87.10	0.503	5.95	0.9996
		Jaccard Diversity	86.99	0.499	6.00	0.9991
	B	Entropy	88.25	0.509	5.88	0.9993
		Margin	86.55	0.519	5.78	0.9992
		Random	86.49	0.502	5.97	0.9996
		Jaccard Diversity	86.46	0.497	6.02	0.9991
	C	Entropy	87.37	0.510	5.87	0.9992
		Margin	84.59	0.517	5.79	0.9992
		Jaccard Diversity	84.34	0.501	5.98	0.9991
		Random	83.26	0.503	5.96	0.9995
30-30-30	A	Entropy	91.68	0.881	3.40	0.9991
		Margin	91.66	0.869	3.45	0.9999
		Random	88.21	0.483	6.20	0.9975
		Jaccard Diversity	88.10	0.446	6.72	0.9944
	B	Entropy	89.30	0.840	3.57	0.9995
		Margin	87.65	0.706	4.24	0.9830
		Random	87.60	0.529	5.66	0.9966

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Table 4 (continued)

Init. Set	Task	Strategy	Plateau (%)	k	x_{95} (%)	R^2
		Jaccard Diversity	87.56	0.530	5.65	0.9958
	C	Entropy	88.42	0.866	3.46	0.9996
		Margin	85.68	0.802	3.74	0.9961
		Jaccard Diversity	85.43	0.673	4.45	0.9981
		Random	84.40	0.251	11.94	0.9974

2.4. Cross-Initial-Set Summary

Table 5 (reproduced from the main paper for convenience) summarises the fitted plateau and x_{95} averaged over Tasks A, B, and C across all three initial sets under Entropy sampling. Fitted plateaus change by less than two points across the three sizes, confirming that achievable F1-Macro is governed primarily by the feature representation and classifier family. The value x_{95} decreases consistently as the initial set grows, from roughly 5.8–5.9% at 5-5-5 to 3.2–3.6% at 30-30-30 for the boosting models.

Table 5. Plateau and x_{95} summary across initial-set sizes, averaged over Tasks A, B, and C (classifier comparison, Entropy sampling).

	5-5-5		**10-10-10**		**30-30-30**	
Classifier	Plateau	x_{95}	Plateau	x_{95}	Plateau	x_{95}
LightGBM	89.50	5.88	89.88	4.08	90.16	3.31
XGBoost	87.89	5.90	88.37	4.02	88.60	3.44
Gradient Boosting	88.33	5.77	88.85	4.15	88.94	3.26
Random Forest	55.64	5.80	56.09	4.18	56.62	3.19
Logistic Regr.	23.97	5.91	24.35	3.99	24.67	3.64

S3. Formal-Execution Cost for Alternative Initial Sets

Table 6 extends the cost analysis in the main paper to the 5-5-5 and 30-30-30 initial sets. The number of new ProB runs added at each checkpoint is identical across initial sets because it depends only on the percentage of the fixed 3 596-system selection pool; only the cumulative run counts change with the initial-set size.

Executing the entire selection pool exhaustively would cost $|\mathcal{M}^F_{\text{pool}}| \times |\mathcal{T}| = 3{,}596 \times 30 = 107{,}880$ ProB runs. At the 4% checkpoint the cumulative cost is at most 7,020 runs (30-30-30 set), representing roughly 6.5% of that exhaustive budget, while the boosting models have already reached near-peak performance (fitted $x_{95} \leq 4.2\%$ for all three initial sets; see Table 3).

S4. Learning-Curve Figures

Figures 1–6 show the classifier and query-strategy learning curves for the 5-5-5 and 30-30-30 initial sets, for all three tasks. Each pair places the classifier comparison (Entropy sampling,

Table 6. Formal-execution cost by checkpoint for the 5-5-5 and 30-30-30 initial sets. Each system requires 30 ProB executions (one per test sequence).

Init. Set	Checkpoint	Added systems	New ProB runs	Total training runs	% of $\mathcal{M}^F_{\text{pool}}$
5-5-5	Initial only	0	0	450	0.0%
	1%	36	1,080	1,530	1.0%
	2%	72	2,160	2,610	2.0%
	4%	144	4,320	4,770	4.0%
	8%	288	8,640	9,090	8.0%
	16%	576	17,280	17,730	16.0%
30-30-30	Initial only	0	0	2,700	0.0%
	1%	36	1,080	3,780	1.0%
	2%	72	2,160	4,860	2.0%
	4%	144	4,320	7,020	4.0%
	8%	288	8,640	11,340	8.0%
	16%	576	17,280	19,980	16.0%

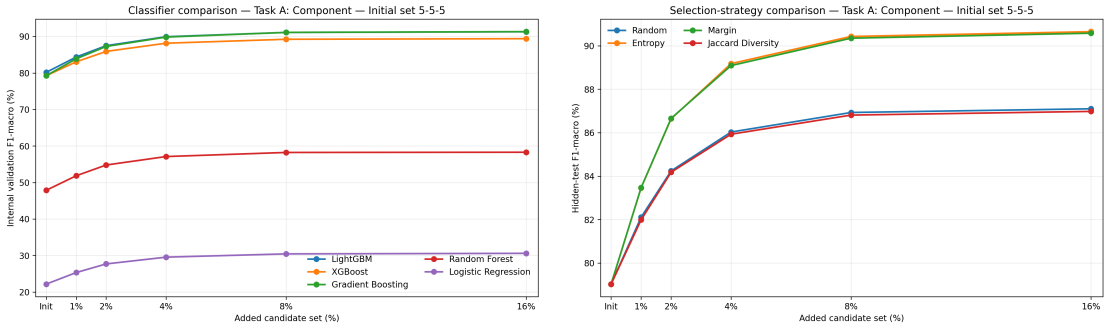


Figure 1. Task A, 5-5-5 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).

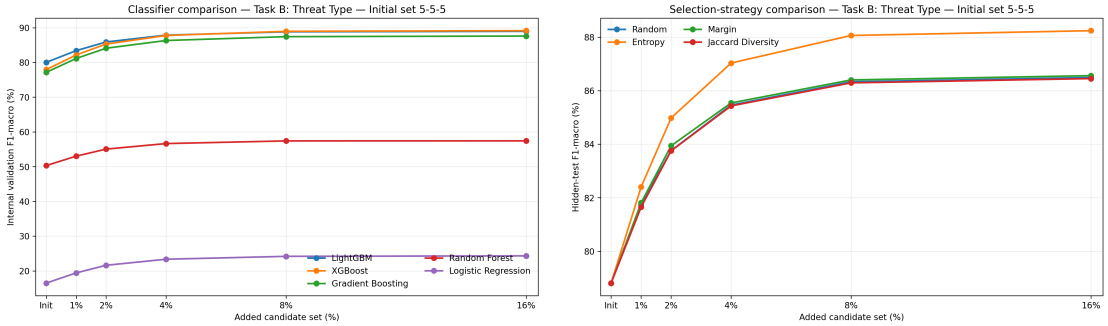


Figure 2. Task B, 5-5-5 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).

five classifiers) beside the query-strategy comparison (LightGBM fixed, four strategies) for the same task and initial set. The trends mirror those of the 10-10-10 results reported in the main paper: the boosting models lead the classifier comparison, and Entropy/Margin lead the strategy comparison.

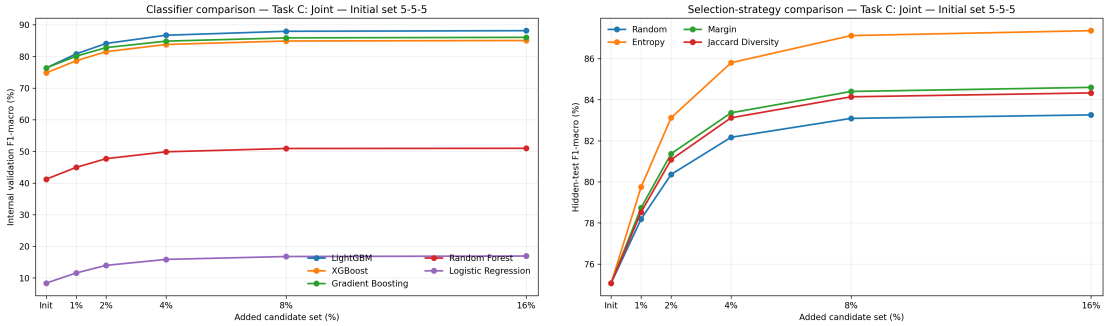


Figure 3. Task C, 5-5-5 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).

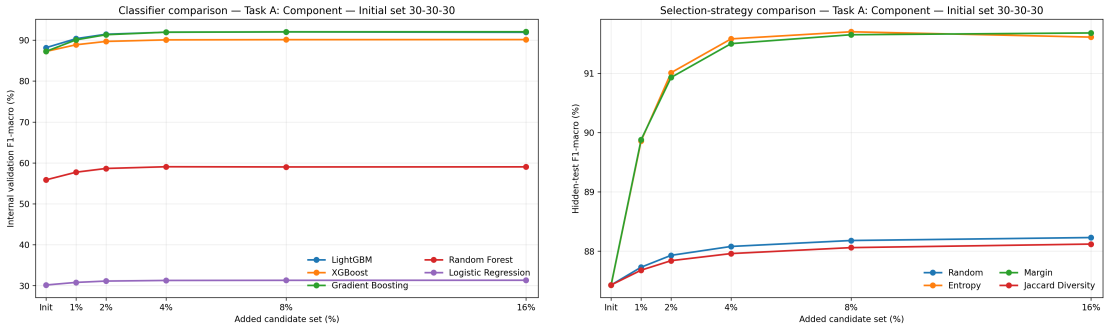


Figure 4. Task A, 30-30-30 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).

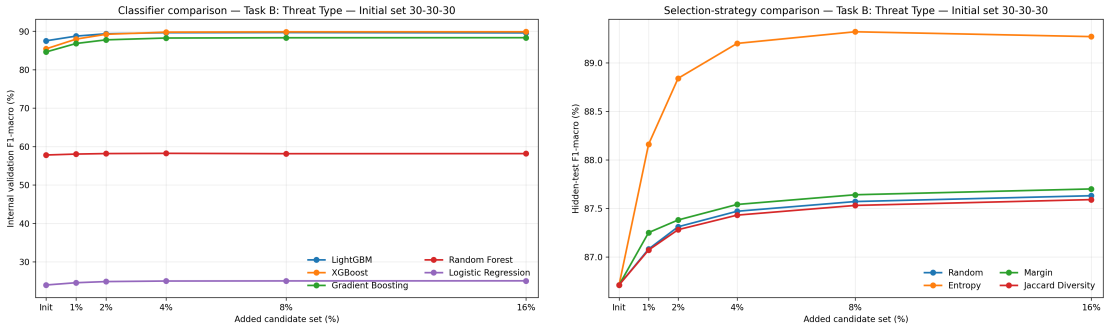


Figure 5. Task B, 30-30-30 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).

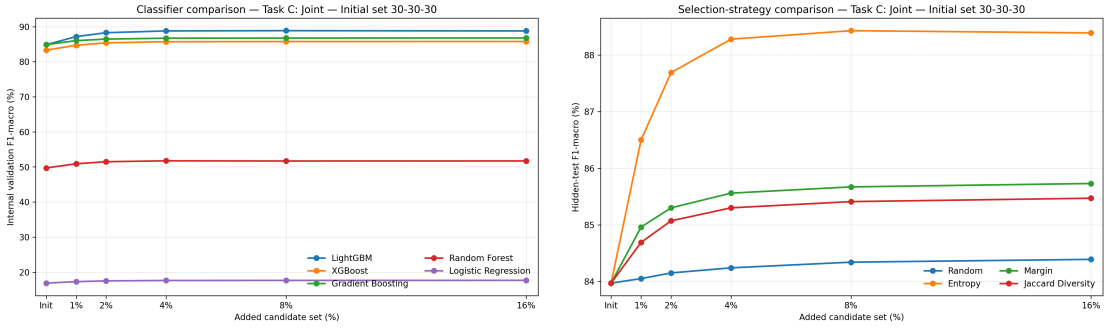


Figure 6. Task C, 30-30-30 initial set. Left: classifier comparison (Entropy sampling). Right: query-strategy comparison (LightGBM).